MIL-T-20616A (Ord)
3 May 1954
SUPERSEDING
MIL-T-20616
31 January 1952

MILITARY SPECIFICATION

TELESCOPE, M86F

1. SCOPE

1.1 This specification covers a telescope used as a sighting device for the 57mm. Recoilless Rifle for direct fire.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and drawing, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

MIL-P-12035 - Packaging of Telescope, M86F

U. S. ARMY

51-7 - Instruments, Fire-Control; General Specification Governing the Manufacture, and Inspection Of.

51-70-2 - Optical Components for Fire-Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection Of.

DRAWING

ORDNANCE CORPS

Class 91, Division 261 - Telescope, M86F

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 General requirements.- General requirements and associated tests common to all fire control instruments as contained in Specifications 51-7 and 51-70-2 shall be mandatory insofar as they are applicable.

3.2 Materials.- Materials shall be in accordance with the applicable drawings, and specifications forming part of this specification.

3.3 Collimation.- The optical axis of the telescope, as determined by the geometric axis of the reticle, shall form an angle of not greater than 2.0 mils with a line which is at right angles to the axis of the conical mounting lug and in a plane passing through the centers of the two mounting lugs.

3.4 Reticle accuracy.- The angular value between the zero range mark and the various range marks shall be as specified on the reticle drawing within 2 percent of the elevation in mils plus 0.4 mil.

3.5 Reticle alignment.- The horizontal reticle line (across the entire field of view) shall be horizontal within an angle subtended by 0.4 mil at one end of the reticle line when the other end is in coincidence with a horizontal target line, when the instrument locating bracket is in a downward vertical position.

3.6 Definition.- Resolving power on the optical axis for a parallel bar test object shall be 11 seconds of arc or better with the entrance pupil restricted to 0.500 inch by means of a suitable aperture stop.

3.7 Parallax.- There shall be no more than 0.2 mil parallax between the image and the reticle at the center of the field when a suitable target is viewed from a distance between 240 yards plus or minus 10 yards.

3.8 Vibration.- After being subjected to the vibration test specified in Specification 51-70-2, except that 2 to 2-1/2 minutes shall be substituted for 15 to 18 seconds, there shall be no damage to the instrument, no change in collimation in excess of 0.2 mil, and no loose dirt in excess of the following:

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3.8.1 Dirt between the adjacent surfaces of the objective lens adjacent to the reticle, both surfaces of the reticle, and the surface of the erector lens adjacent to the reticle shall not exceed one in number (on each surface) of a size not larger than the width of a reticle line.

3.8.2 Dirt between the adjacent surfaces of the erector lens and the surface of the field lens adjacent to the erector lens shall not exceed 3 in number on each surface of a size not larger than a 50 dig.

3.8.3 Dirt between the adjacent surfaces of the field lenses shall not exceed 3 in number of a size not larger than a 40 dig.

3.8.4 Dirt between the adjacent surfaces of the field lens and eyelens shall not exceed 3 in number of a size not larger than a 20 dig.

3.9 Sealing.- The telescope shall show no evidence of leakage when subjected to the sealing test.

3.10 Temperature.- The temperature requirements and test Type II specified in Specification 51-7 shall apply, except minus 80°F shall be substituted for minus 60°F.

3.11 Illumination.- The reticle markings shall appear clearly defined when observed in a dark room utilizing Instrument Light M45 or equivalent.

3.12 Workmanship.- Workmanship shall be in accordance with Specifications 51-7 and 51-70-2.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified, each telescope shall be inspected for conformance with this specification and as specified in Specifications 51-7 and 51-70-2.

4.2 Definition test.- The test object shall be of the pattern illustrated in Specification 51-70-2, figure VI-A. The image shall be viewed with the aid of a six power auxiliary telescope of improved design, scaled in diopters. The test pattern shall be resolved in each of the four meridians by an adjustment of the auxiliary telescope not to exceed plus or minus 1/8 diopter from the setting for best focus of the

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numeral in the center of the test pattern. Illumination of the test chart shall be such that the image seen through the auxiliary telescope will have a minimum apparent brightness of 10 millilamberts.

4.3 Sealing test.- Sealing test shall be in accordance with procedure and test Drawing F7680673, Fixture Pressure Testing.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing, and marking shall be in accordance with Specification MIL-P-12035.

6. NOTES

6.1 Ordering data.- Invitation for bids and contracts should specify the title, number, date of this specification and type of packing.

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